Homework #9

This is the homework for week 10. The due date is 11:59 p.m., November 26. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (35 points) Suppose data $X_1, \cdots, X_n$ are i.i.d. and satisfy Poission distribution, i.e.,

$$Prob(X_i = k) = \frac{\lambda^k}{k!} \exp(-\lambda).$$

Suppose we want to test the hypothesis

$$H_0 : \lambda = \lambda_0 \text{ vs } H_1 : \lambda \neq \lambda_0.$$

i. Suppose we want to use likelihood ratio test. Derive the formula of the test statistics and the p-value.

ii. Suppose we want to use Wald test. Derive the formula of the test statistics and the p-value.

iii. Suppose we want to use Score test. Derive the formula of the test statistics and the p-value.

iv. Suppose we have data set $[4, 1, 1, 2, 1, 3, 1, 3, 4, 3, 4, 0, 2, 2, 2, 1, 0, 1, 2, 3]$, and $\lambda_0 = 1.0$. For these three tests, with size (Type-I error) 5%, do we reject H_0?

Solution:

i. The log-likelihood function is

$$\ell(\lambda) = (\sum_{i=1}^{n} X_i) \log(\lambda) - n\lambda \Rightarrow \widehat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Therefore, the test-statistics is

$$\widehat{T} = 2 \sum_{i=1}^{n} X_i (\log(\widehat{\lambda}) - \log(\lambda_0)) - 2n(\widehat{\lambda} - \lambda_0)$$

and the P-value is

$$\widehat{p} = 1 - \Psi(\widehat{T}),$$

where Ψ is the cdf of χ_1^2 distribution.

ii. The Fisher information is given by

$$I(\lambda) = -\mathbf{E}\ell''(\lambda) = \frac{n}{\lambda},$$

so the Wald test statistics and the P-value is given by

$$\widehat{W} = \frac{n}{\widehat{\lambda}}(\widehat{\lambda} - \lambda_0)^2, \widehat{p} = 1 - \Psi(\widehat{W}).$$

(Notably, this is also walid

$$\widehat{W} = \widehat{I}(\widehat{\lambda} - \lambda_0)^2, \text{ where } \widehat{I} = \frac{\sum_{i=1}^n X_i}{\widehat{\lambda}^2}$$

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iii. The Score function is

$$U(\lambda) = \frac{\sum_{i=1}^n X_i}{\lambda} - n,$$

so the test statistics and the P-value are

$$\widehat{U} = \lambda_0 \left(\frac{\sum_{i=1}^n X_i}{\lambda_0} - n \right)^2 / n, \ \widehat{p} = 1 - \Psi(\widehat{U}).$$

iv. For this dataset, $\sum_{i=1}^n X_i = 40$ and $\widehat{\lambda} = 2.0$. Therefore, $\widehat{T} = 15.45$, $\widehat{W} = 10.0$, $\widehat{U} = 20.0$, which is greater than 3.84, so we reject H_0.

(**Question2):** (35 points) Let $X_1, \cdots, X_n$ be n independent random variables with normal distributions $X_i \sim N(x_i\beta, \sigma^2)$, where x_i and σ^2 are known. We want to test the hypothesis

$$H_0 : \beta = \beta_0 \ vs \ H_1 : \beta \neq \beta_0.$$

i.Suppose we want to use likelihood ratio test. Derive the formula of test statistics and the P-value of the test.

ii. Suppose we want to use Wald test. Derive the formula of test statistics and the P-value of the test.

The log-likelihood function is

$$\ell(\beta) = -\frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - x_i\beta)^2 - \frac{n}{2}\log(2\pi\sigma^2) \Rightarrow \ell'(\beta) = \frac{1}{\sigma^2}\sum_{i=1}^n (X_i - x_i\beta) \Rightarrow \widehat{\beta} = \frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n x_i}.$$

So the test statistics and the P-value are

$$\widehat{T} = \frac{1}{\sigma^2}\sum_{i=1}^n (X_i - x_i\beta_0)^2 - \frac{1}{\sigma^2}\sum_{i=1}^n (X_i - x_i\beta)^2, \widehat{p} = 1 - \Psi(\widehat{T}).$$

ii. For the Fisher information is

$$I(\beta) = -\mathbf{E}\ell''(\beta) = \frac{\sum_{i=1}^n x_i}{\sigma^2},$$

the Wald test statistics and P-value are given by

$$\widehat{W} = \frac{\sum_{i=1}^n x_i}{\sigma^2}(\widehat{\beta} - \beta_0)^2, \ \widehat{p} = 1 - \Psi(\widehat{W}).$$

2 Programming Questions

(30 points) In this assignment, we will investigate the performance of P-value under H_0 and H_1. Let us suppose observations $X_1, \cdots, X_n \sim N(\mu, \sigma^2)$, with σ^2 known. Suppose we want to test the hypothesis

$$H_0 : \mu = \mu_0 \text{ versus } H_1 : \mu > \mu_0.$$

i. Suppose $\mu = \mu_0 = 1.0$, $\sigma^2 = 2$, $n = 100$. Write down the formula of P-value.

ii. Generate a set of data $X_1, \cdots, X_{100} \sim N(1, 2)$. Calculate the P-value p_i using your formula. Repeat this for 500 times, and derive the p-values $p_1, \cdots, p_{500}$. Plot a histogram of p_i, and it should look like uniform distribution.

iii. Now suppose $\mu = 1.0$ and $\mu_0 = -1.0$. Again calculate the P-value p_i and repeat for 500 times. Plot a histogram of p_i, and the p-value should concentrate on 0.